

Feedforward Filters

General Form $y_t = x_t + ax_{t-1}$ $H(\omega) = 1 + ae^{-i\omega}$ $H(z) = 1 + az^{-1}$ $ H(\omega) = (1 + a^2 + 2a \cos(\omega))^{\frac{1}{2}}$	Transfer Function $y_t = x_t + ax_{t-1}$ $\mathbf{Y} = \mathbf{X}(1 + az^{-1})$ $\frac{\mathbf{Y}}{\mathbf{X}} = (1 + az^{-1})$ $H(z) = 1 + az^{-1}$...
Inverse Comb $y_t = x_t + R^L x_{t-L}$ $H(\omega) = 1 + R^L e^{-i\omega L}$ $H(z) = 1 + R^L z^{-L} = \frac{z^L - R^L}{z^L}$ $ H = 2 \sin(\omega L/2) $	Zeroes $z^L = R^L$ Poles $z = 0$ $z = Re^{i\theta}$ $e^{iL\theta} = 1$ $\frac{L\theta}{2\pi} = k$
Phase Response $[R_\omega e^{i\theta_\omega}]$ R_ω : MAGNITUDE RESPONSE $\theta_\omega = \tan^{-1}(\frac{ImaginaryPart}{RealPart})$ $\theta_\omega H(\omega) = e^{i\omega}(1 + ae^{-i\omega})$ $e^{i\theta\omega} = \frac{H(\omega)}{ H(\omega) }$	
$y_t = x_t - 2x_{t-1}$ $H(\omega) = 1 - 2e^{-i\omega} = \frac{e^{i\omega} - 2}{e^{i\omega}}$	$ H(\omega) = [5 - 2\cos(\omega)]^{1/2}$ $\theta_\omega = \tan^{-1}(\frac{2\sin(\omega)}{1 - 2\cos(\omega)})$
Palindromic Phase Response ODD TERMS $H(z) = 2 + z^{-1} - 3z^{-2} + z^{-3} + 2z^{-4}$ $H(\omega) = 2 + e^{-i\omega} - 3e^{-2i\omega} + e^{-3i\omega} + 2e^{-4i\omega}$ $H(\omega) = e^{-i2\omega}[4\cos(2\omega) + 2\cos(\omega) - 3]$ $ H(\omega) = 4\cos(2\omega) + 2\cos(\omega) - 3$ $\theta_\omega = -2\omega$	
Even Terms $y_t = x_t + \frac{1}{3}x_{t-1} + \frac{1}{3}x_{t-2} + x_{t-3}$ $H(\omega) = 1 + \frac{1}{3}e^{-i\omega} + \frac{1}{3}e^{-2i\omega} + e^{-3i\omega}$ $H(\omega) = e^{-3i\frac{\omega}{2}}[e^{3i\frac{\omega}{2}} + \frac{1}{3}e^{i\frac{\omega}{2}} + \frac{1}{3}e^{-i\frac{\omega}{2}} + e^{-3i\frac{\omega}{2}}]$	
Collect Cos with $e^{i\frac{\omega}{2}} + e^{-i\frac{\omega}{2}} = 2\cos(\frac{\omega}{2})$	
Phasors $f(t) = e^{i\omega t}$ ω = angular freq (rad/s) Frequency: $f = \frac{\omega}{2\pi}$ Hz	Period: $T = \frac{2\pi}{\omega} = \frac{1}{f}$ sec Function $f(z) = \alpha z$ rotates by arg(α), scales by $ \alpha $
Aliasing Sample rate: f_s Hz Nyquist frequency: $f_s/2$	$f_{alias} = f - nf_s $ (choose n to minimize)

Feedback Filters

General Form $y_t = x_t + Ry_{t-1}$ $ H(\omega) = \frac{1}{ 1 - Re^{-i\omega} }$	$H(z) = \frac{1}{1 - Rz^{-1}} = \frac{z}{z - R}$ Pole at $z = R$ (stable if $ R < 1$)
Reson Filter (2-Pole) $y_t = A_0x_t + 2R \cos \theta \cdot y_{t-1} - R^2y_{t-2}$ $H(z) = \frac{1}{(z - Re^{i\theta})(z - Re^{-i\theta})}$ $ H(\omega) \approx \frac{1}{\sqrt{1 - R^2}}$ at resonance Poles on unit circle at $z = Re^{\pm i\theta}$	
Gain factor $A_0 = (1 - R^2)\sin\theta$ Pole angle $\theta = \frac{2\pi\psi}{fs}$ $\cos(\psi) = \frac{1 + R^2}{2R} \cos(\theta)$	... Pole Radius $R \approx 1 - B/2$ Bandwidth $B \approx 2(1 - R)$
Plucked String Filter $y_t + ay_{t-1} = ax_t + x_{t-1}$ $H(z) = \frac{a + z^{-1}}{1 + az^{-1}}$ $ H(\omega) \rightarrow const$ (phase does not affect magnitude)	
Delay $L = \lfloor P \rfloor$ Period $P = \frac{fs}{f_0}$ $\frac{fs}{f} = L + \frac{1}{2} + \delta$	Low-pass $y = \frac{1}{2}(x_t + x_{t-1})$ $[\frac{1}{2}delay]$
	All-Pass $a = \frac{1 - \delta}{1 + \delta}$ $\theta_\omega = 2\theta$ $\tan^{-1}x \approx x$ $\theta_\omega = 2(\frac{a-1}{a+1}\omega)$
	Phase Delay $\delta = \frac{2}{\omega} \tan^{-1}[\frac{a-1}{a+1} \tan(\omega/2)]$ $\tau(\omega) = \frac{-2 \arctan(\frac{a \sin \omega}{1 + a \cos \omega})}{\omega}$
Comb Filter [HARMONIC SERIES] $H(z) = \frac{1}{1 - R^L z^{-L}}$ $ H = 2 \cos(\omega L/2) $	
Peaks $\omega = 2\pi k/L$ Zeroes $1 - R^L z^{-L} = 0$	Poles $-1 = e^{i\pi}$ $z^L = -1$
Impulse Response Sum $\sum_{t=0}^{\infty} h_t = \frac{1}{1 - R^L}$	

Periodic Sound

TIME DOMAIN SIGNALS CONSIDERED TO BE PERIODIC FUNCTIONS $f(t)$ TIME DOMAIN $\neq$ FREQUENCY DOMAIN	
$\omega_0 = \frac{2\pi}{T}$ $f(t) = e^{ik\omega_0 t}$ $\int e^{at} = \frac{1}{a}e^{at} + C$	$\int e^{iat} = \frac{1}{ia}e^{iat}$ $f(t + kT) + f(t)$ $\frac{1}{i} = -i$ $\frac{1}{-i} = i$
Inner Product $\frac{1}{T} \int_0^T f(t)g(t)dt$ $\langle u, v \rangle = \sum_n \bar{u}_n v_n$...	
Forward Fourier $C_k = \frac{1}{T} \int_0^T f(t)e^{ik\omega_0 t}$...	
Inverse Fourier $f(t) = C_0 + 2 \sum_{k=1}^{\infty} [a_k \cos(k\omega_0 t) - b_k \sin(k\omega_0 t)]$	
DFT & Fourier Basis Basis Vectors: $u_k[n] = e^{i2\pi kn/N}$ Orthogonality: $u_k \cdot u_j = N\delta_{kj}$	
$X[k] = \sum_{n=0}^{N-1} x[n]e^{-i2\pi kn/N}$	
$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k]e^{i2\pi kn/N}$	
Coefficient: $c_k = X[k]/N$	
Square Wave Fourier Period T , 50% duty cycle:	
$c_k = \begin{cases} 0 & k \text{ even} \\ \frac{4}{k\pi} & k \text{ odd} \end{cases}$	
General Fourier: $c_k = \frac{1}{T} \int_0^T s(t)e^{-i2\pi kt/T} dt$	
Calculations Quantization ROOT-MEAN-SQUARE, SIGNAL-TO-NOISE RATIO	
RMS Error $[a, b] \rightarrow \frac{1}{b-a} \int_a^b f(x)dx$ $[0, \frac{1}{2}] \rightarrow \frac{1}{0.5} \int_0^{0.5} x^2 dx = \frac{1}{\sqrt{12}}$	SNR $-2^{B-1} \leq x_t \leq 2^{B-1} - 1$ $2(2^{B-1}) = 2^B$ $\frac{2^{B-1}}{RMS} = 2^B \sqrt{3}$
$[-0.5, 0.5] \rightarrow \frac{1}{2\sqrt{3}} \approx 0.289$	
dB = $20 \log_{10}(A)$	

Unit Circle $e^{i\pi/2} = i, e^{i\pi} = -1,$ $e^{i3\pi/2} = -i, e^{i2\pi} = 1$ $e^{i\pi/3} = \frac{1}{2} + \frac{\sqrt{3}}{2}i,$ $e^{i2\pi/3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ $e^{i\pi/4} = \frac{1+i}{\sqrt{2}},$ $e^{i\pi/6} = \frac{\sqrt{3}}{2} + \frac{1}{2}i$	Complex Arithmetic $z = a + bi = re^{i\theta},$ $r = z = \sqrt{a^2 + b^2}$ $i^2 = -1, i^3 = -i$ $i^4 = 1, i^5 = i$								
Operations $(a + bi)(c + di) = (ac - bd) + (ad + bc)i$ $\frac{a+bi}{c+di} = \frac{(a+bi)(c-bi)}{c^2+d^2} = \frac{(ac+bd)+(bc-ad)i}{c^2+d^2}$ $(re^{i\theta})^n = r^n e^{in\theta}$ $(a + bi)(a - bi) = a^2 + b^2$ Conjugate: $\bar{z} = a - bi = re^{-i\theta}$ Rotation by θ : multiply by $e^{i\theta}$	Trig Identities $1 + e^{i\omega} = 2 \cos(\omega/2)$ $1 - e^{i\omega} = 2i \sin(\omega/2)$ $e^{it} - e^{-it} = 2i \sin(t)$ $e^{it} + e^{-it} = 2 \cos(t)$ $\sin(2a) = 2\sin(a)\cos(a)$								
	... $\cos(2a) = \cos^2(a) - \sin^2(a)$ $\sin^2(a) = \frac{1 - \cos(2a)}{2}$ $\cos^2(a) = \frac{1 + \cos(2a)}{2}$								
	Quadratic from Complex Roots $(x - (a + bi))(x - (a - bi)) = x^2 - 2ax + (a^2 + b^2)$ Cubic Factoring $x^3 + 1 = (x + 1)(x^2 - x + 1)$ $x^3 - 1 = (x - 1)(x^2 + x + 1)$								
Taylor Series $e^{ix} = 1 + i\frac{x}{1} + \frac{i^2(\frac{x}{2})^2}{2!} + \dots$	Product $\prod_{k=1}^n e^{ik\theta} = e^{i\theta n(n+1)/2}$ Geometric Series $\sum_{k=0}^n e^{ik\theta} = \frac{1 - (e^{i\theta})^{n+1}}{1 - e^{i\theta}}$								
	+3dB = $\sqrt{2}$ × amplitude +6dB = 2× amplitude +20dB = 10× amplitude								
Filter Combinations • Cascade: $H_{total} = H_1 \cdot H_2$ • Parallel: $H_{total} = H_1 + H_2$ • Feedback: $H_{total} = \frac{GH}{1 - FH}$									
Phase Properties Linear Phase: • Symmetric coefficients • $\phi(\omega) = -\omega D$ where D = group delay • No distortion of wave shape Phase Delay vs Group Delay: • Phase delay: $\tau_p = -\phi(\omega)/\omega$ • Group delay: $\tau_g = -d\phi/d\omega$									
<table><tr><th>Operation</th><th>Formula</th></tr><tr><td>Complex dot</td><td>$(a_1, b_1) \cdot (a_2, b_2) = \bar{a}_1 a_2 + b_1 b_2$</td></tr><tr><td>Parseval</td><td>$\frac{1}{T} \int s ^2 dt = \sum c_k ^2$</td></tr><tr><td>Impulse</td><td>$\delta[n] = \{1 \text{ if } n = 0, 0 \text{ else}\}$</td></tr></table>	Operation	Formula	Complex dot	$(a_1, b_1) \cdot (a_2, b_2) = \bar{a}_1 a_2 + b_1 b_2$	Parseval	$\frac{1}{T} \int s ^2 dt = \sum c_k ^2$	Impulse	$\delta[n] = \{1 \text{ if } n = 0, 0 \text{ else}\}$	
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Polynomial Factoring

Factor $x^3 + 1$ and find all roots

Step 1: Use sum of cubes formula

- $x^3 + 1 = (x + 1)(x^2 - x + 1)$
- First root: $x = -1$

Step 2: Quadratic formula on $x^2 - x + 1 = 0$

- $x = \frac{1 \pm \sqrt{1-4}}{2} = \frac{1 \pm \sqrt{-3}}{2}$
- $= \frac{1 \pm i\sqrt{3}}{2}$

Step 3: All three roots

- Root 1: $x = -1$
- Root 2: $x = \frac{1+i\sqrt{3}}{2} = e^{i\pi/3}$
- Root 3: $x = \frac{1-i\sqrt{3}}{2} = e^{-i\pi/3}$
- These are the three cube roots of -1

Aliasing

Find alias: $f_s = 44100$ Hz, signal = 32100 Hz

Step 1: Check Nyquist

- Nyquist freq = $f_s/2 = 22050$ Hz
- $32100 > 22050$, so aliasing occurs

Step 2: Calculate alias frequency

- Use: $f_{alias} = |f - nf_s|$ where n minimizes
- $n = 0$: $|32100 - 0| = 32100$ Hz
- $n = 1$: $|32100 - 44100| = 12000$ Hz (smaller!)
- $n = 2$: $|32100 - 88200| = 56100$ Hz (larger)

Step 3: Verify and conclude

- 12000 Hz is in valid range $[0, 22050]$
- The 32100 Hz signal appears as 12000 Hz

DFT Calculation

Find $X[1]$ for $x = [1, 0, 1]$, $N = 3$

Step 1: Write DFT formula

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-i2\pi kn/N}$$

Step 2: Substitute values ($k = 1$, $N = 3$)

- $X[1] = x[0]e^0 + x[1]e^{-i2\pi/3} + x[2]e^{-i4\pi/3}$
- $= 1(1) + 0(e^{-i2\pi/3}) + 1(e^{-i4\pi/3})$
- $= 1 + e^{-i4\pi/3}$

Step 3: Simplify the exponential

- $e^{-i4\pi/3} = \cos(-4\pi/3) + i\sin(-4\pi/3)$
- $= \cos(4\pi/3) - i\sin(4\pi/3)$
- $= -1/2 - i \cdot (-\sqrt{3}/2) = -1/2 + i\sqrt{3}/2$

Step 4: Final answer

- $X[1] = 1 + (-1/2 + i\sqrt{3}/2) = 1/2 + i\sqrt{3}/2$
- This equals $e^{i\pi/3}$
- DFT coefficient: $c_1 = X[1]/N =$

FIR Filter Analysis

Analyze $y_t = x_t - (1/4)x_{t-1}$

Step 1: Transfer function

- Direct: $H(z) = 1 - (1/4)z^{-1}$

Step 2: Find zeros and poles

- Zero: Set $1 - (1/4)z^{-1} = 0$
- Solve: $z^{-1} = 4$, so $z = 1/4$
- Pole: At $z = 0$ (from z^{-1} term)
- Zero at $z = 1/4$, pole at $z = 0$

Step 3: Frequency response at $\omega = 0$

- $H(0) = 1 - (1/4)e^{-i \cdot 0} = 1 - 1/4 = 3/4$
- DC gain is 0.75 (attenuation)

Step 4: Magnitude at $\omega = \pi/2$

- $H(\pi/2) = 1 - (1/4)e^{-i\pi/2}$
- $e^{-i\pi/2} = -i$, so $H(\pi/2) = 1 - (1/4)(-i) = 1 + i/4$
- $|H(\pi/2)| = |1 + i/4| = \sqrt{1 + 1/16} =$

IIR Filter with Impulse

Analyze $y_t = x_t - (1/4)y_{t-1}$, impulse response

Step 1: Find transfer function

- Rearrange: $y_t + (1/4)y_{t-1} = x_t$
- Z-transform: $Y(z)[1 + (1/4)z^{-1}] = X(z)$
- $H(z) = 1/[1 + (1/4)z^{-1}] = z/(z + 1/4)$

Step 2: Check stability

- Pole at $z = -1/4$
- $|pole| = |-1/4| = 1/4 < 1$ STABLE

Step 3: Impulse response calculation

- Input: $x = [1, 0, 0, \dots]$

- $y_0 = 1 - (1/4)(0) = 1$
- $y_1 = 0 - (1/4)(1) = -1/4$
- $y_2 = 0 - (1/4)(-1/4) = 1/16$
- $y_3 = 0 - (1/4)(1/16) = -1/64$

Step 4: Pattern recognition

- $h[n] = (-1/4)^n$ for $n \geq 0$
- Alternating sign, exponential decay
- Sum: $\sum h[n] = 1/(1 - (-1/4)) = 4/5$

Reson Filter Design

Design reson: $\omega = \pi/4$, bandwidth $B = 0.1$

Step 1: Calculate pole radius from bandwidth

- Narrow band formula: $B \approx 2(1 - R)$
- $0.1 = 2(1 - R)$, so $1 - R = 0.05$
- $R = 0.95$

Step 2: Place complex conjugate poles

- Poles at $z = 0.95e^{\pm i\pi/4}$
- In rectangular: $0.95 \cdot (\frac{1}{\sqrt{2}} \pm i\frac{1}{\sqrt{2}})$
- $= \frac{0.95}{\sqrt{2}}(1 \pm i) = 0.672(1 \pm i)$

Step 3: Write transfer function

- $H(z) = \frac{1}{1 - 2R\cos(\pi/4)z^{-1} + R^2z^{-2}}$
- $\cos(\pi/4) = 1/\sqrt{2}$, so $2R\cos(\pi/4) = 2(0.95)/\sqrt{2} = 1.343$
- $H(z) = \frac{1}{1 - 1.343z^{-1} + 0.9025z^{-2}}$

Step 4: Calculate peak gain

- At resonance ($\omega = \pi/4$):
- Approximate gain $\approx 1/(1 - R) = 1/0.05 = 20$
- In dB: $20\log_{10}(20) = 26$ dB

Plucked String Tuning

Tune to 440 Hz, $f_s = 44100$ Hz

Step 1: Calculate ideal period in samples

- $P = f_s/f_0 = 44100/440 = 100.227$ samples
- Need to achieve this total delay

Step 2: Allocate delays

- Try integer delay $L = 100$
- Low-pass filter adds 0.5 samples
- Total so far: $100 + 0.5 = 100.5$

- Still need: $100.227 - 100.5 = -0.273$ (negative!)

Step 3: Adjust allocation

- Use $L = 99$ instead
- With LP: $99 + 0.5 = 99.5$
- Need from all-pass: $100.227 - 99.5 = 0.727$

Step 4: Calculate all-pass coefficient

- For delay $\delta = 0.727$:
- $a = (1 - \delta)/(1 + \delta) = 0.273/1.727 = 0.158$

Step 5: Verify total delay

- Integer: 99 samples
- Low-pass: 0.5 samples
- All-pass: 0.727 samples
- Total: $99 + 0.5 + 0.727 = 100.227$ Perfect!

All-Pass Identification

Is $H(z) = (0.5z + 1)/(z + 0.5)$ all-pass?

Step 1: Convert to standard form

- Divide num and denom by z :
- $H(z) = (0.5 + z^{-1})/(1 + 0.5z^{-1})$

Step 2: Check all-pass structure

- All-pass form: $H(z) = (a + z^{-1})/(1 + az^{-1})$
- Our filter: numerator has $a = 0.5$
- Denominator also has $a = 0.5$
- Structure matches! YES, all-pass

Step 3: Verify magnitude = 1

- At any frequency: $|H(e^{i\omega})| = |(0.5 + e^{-i\omega})/(1 + 0.5e^{-i\omega})|$
- Numerator and denominator have same structure
- Can show $|H(e^{i\omega})| = 1$ for all ω

Step 4: Phase delay property

- All-pass changes phase without changing magnitude
- Used for fractional delay adjustment
- Phase: $\phi(\omega) = -2\arctan(\frac{0.5\sin\omega}{1+0.5\cos\omega})$